

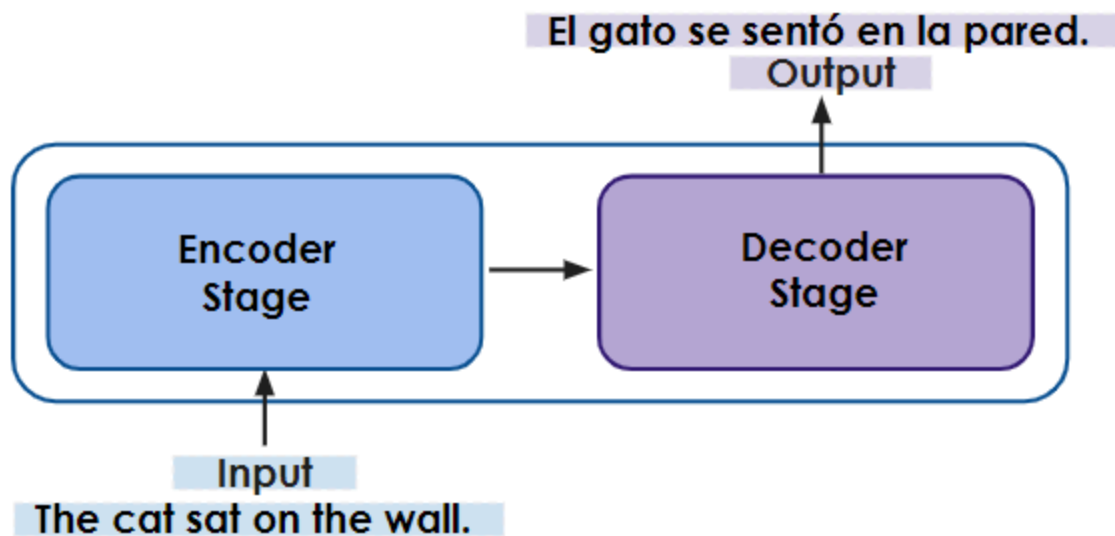
Tokenization

A token is a basic unit of language, representing individual elements such as words, sub words, or characters. Tokenization is the process of breaking down a text or a sequence of words into these smaller units.

Tokenization also helps in understanding the context of a sequence. By breaking down text into tokens, NLP models can capture the sequential relationships and dependencies between words. It enables the model to discern the order of words, which is crucial for tasks like language understanding, sentiment analysis, and part-of-speech tagging.

Encoder-Decoder

In the field of AI and machine learning, the encoder-decoder architecture stands as a pivotal framework, prominently employed for tasks requiring sequence-to-sequence modeling in natural language processing (NLP). This versatile architecture is extensively utilized in applications such as language translation, text summarization, and question-answering systems. The two-stage process involves encoding input data into a fixed-length numerical representation through an encoder and decoding it to generate the desired output format using a decoder.



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Components of Encoder-Decoder Architecture:

Encoder: The encoder transforms input sequences into a fixed-length vector, often referred to as a hidden or latent representation. This condensed representation captures essential information from the input sequence.

Various neural network architectures, including RNNs, LSTMs, CNNs, and transformers, can serve as the foundation for the encoder.

Decoder: The decoder generates an output sequence based on the latent representation obtained from the encoder.

